

[Answer any four questions taking two from each group]

Group A

- Q1 a) Determine whether following statements are true or false and explain your answer (any six): 6
- Interpreter translates the entire program.
 - Super class variable cannot reference subclass object.
 - In Java, *main()* function must be resided in a *public* class.
 - Signature of a method must be same in method overloading.
 - Return type of all the functions must be different in case of method overloading.
 - Finalize method is called when an object is out of scope.
 - Wrapper classes allow primitive types to be accessed as objects.
 - If an object is visible to more than one thread, all reads or writes to that object's variables are done through synchronized methods.

- b) Describe Erasure. Why is Java called the "Platform Independent Programming Language"? 3+3

- c) Create a class *Account* with two overloaded constructors. The first constructor is used for initializing, the name of account holder, the account number and the initial amount in the account. The second constructor is used for initializing the name of the account holder, the account number, the addresses, the type of account and the current balance. The *Account* class will have the methods *deposit()*, *withdraw()*, and *getBalance()*. Make the necessary assumptions for data members and return type of the methods. Create objects of *Account* class and use them. 6

- d) What are the restrictions of *generics*? How *generics* improve type safety? 3+4

- Q2 a) Describe constructor overloading with example. Describe the circumstances in which an abstract method would be appropriate. 3+3
- b) Check the following java program carefully. Do you think there is any compilation error? If there is find the errors, correct it using line number and then write the output. For correcting you cannot insert new statement or delete a statement. If there is no error, then write only output. 12

```
1. class BlockExam {  
2. { System.out.println("B1: Exam Started!"); }  
3. int a; static int s = 33; ✓  
4. BlockExam(int a) { this.a = a;  
5. System.out.println("B2: Inside Constructor"); }  
6. a = 11; s = 22;  
7. System.out.println("B3: a = " + a); }  
8. void m() { System.out.println("M1: a = " + a); }  
9. static { System.out.println("B4: a = " + a + ", s = " + s); }  
10. static void sm(int a, int s) {  
11. System.out.println("M2: a = " + a + ", s = " + BlockExam.s);  
12. BlockExam.s = a + s;  
13. }  
14. { s += 44; }  
15. static { System.out.println("B5: s = " + s); }  
16. public static void main(String[] args) {  
17. System.out.println("Main Started!");  
18. BlockExam.sm(20, 10); ✓  
19. BlockExam b = new BlockExam(30);  
20. b.m(); b.sm(50, 40);  
21. }  
22. System.out.println("B6: Exam Ended!"); }  
23. }
```

- c) At what point is the Bar object, created on line 6, eligible for garbage collection? 7

```
1. class Bar {  
2. }  
3.  
4. public class Test {  
5. Bar doBar() {  
6. Bar b = new Bar();  
7. return b;  
8. }  
9. public static void main (String[] args) {  
10. Test t = new Test();  
11. Bar newBar = t.doBar();  
12. System.out.println("newBar");  
13. newBar = new Bar(); ✓  
14. System.out.println("finishing");  
15. }  
16. }
```

- Q3 a) Show the use of "this" and "super" keywords.
 b) What is Final Class? What is the purpose of declaring a variable or method as final?
 c) Differentiate among abstract class, generic class and interface.
 d) Check the following code carefully:

<pre>public class Person { public void work() { System.out.println("Working"); } public void sleep() { System.out.println("Sleeping"); } }</pre>	<pre>public class Teacher extends Person { public void work() { System.out.println("Teaching"); } public void work(String course) { System.out.println("Teaching course : " + course); } }</pre>
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Now write the output of the following codes with explanation to your answer. If there is any error, specify it with reason.

- i. `Person p = new Person();`
- ii. `p.work();`
- iii. `p.sleep();`
- iv. `p.work("OOP");`
- v. `Teacher t = new Teacher();`
- vi. `t.work("OOP");`
- vii. `t.sleep();`
- viii. `t.work();`

Group B

- Q4 a) Define True/False. If false give correct answer (any six).
- i. Java supports implementation of multiple interface in a single class.
 - ii. Methods that are declared as final cannot be overridden.
 - iii. An abstract class can inherit an interface.
 - iv. Interfaces cannot have instance variables.
 - v. In Java, class names must start with an uppercase letter.
 - vi. The type parameters in generic class can be used to declare the return type of a method.
 - vii. `wait()` causes current thread to wait until another thread invokes the `notify()` method.
- b) Check the following java program carefully. There are several compilation errors. Find the errors and specify the reason behind it, then provide possible correction using line number. For correcting you cannot insert new statement or delete a statement.

```
1. interface A {
2.     int a = 10; final int b = 20;
3.     void am();
4. }
5. interface B extends A {
6.     static int b = 30; int c = 50;
7.     interface C {
8.         final static int c = 40;
9.         void bm();
10.        void am();
11.    }
12.    void bm();
13. }
14. class InterfaceExam implements B, C {
15.    int a = 60;
16.    public static void main(String[] args){
17.        InterfaceExam inf = new InterfaceExam();
18.        inf.am();
19.        A a = inf; B b = inf; C c = inf;
20.        a.am(); b.am(); b.bm(); c.bm();
21.    }
22.    void am() { System.out.println("AM: = " + c); }
23.    void bm() { System.out.println("BM: = " + a); }
24. }
```

- c) Is the following code correct or incorrect? Justify your answer. If incorrect, specify why and rewrite the correct code. 3+3

```

1. public abstract class A {
2.     final abstract void show();
3. }
4. public class B extends A{
5.     void show(int a){}
6. }

```

- d) What happens when the constructor of a class is declared as "private"? Write down the difference between "protected" and "default" access specifiers. 2+3

- Q5 a) Can you override private or static method in Java? If possible show an example, if not write the reason. 2+2

- b) Briefly describe Exception Handling with appropriate examples. 5

- c) Define the class Point, Shape, Rectangle and Circle for the following program. All member variables of these classes should be private. 4*4

```

1. public class ObjectToClass {
2.     public static void main(String[] args) {
3.         Point p = new Point(11, 22);
4.         System.out.println("Point: " + p.getX() + ", " + p.getY());
5.         Rectangle r = new Rectangle(p, 30.0, 50.0);
6.         System.out.println("R: " + r.getHeight() + ", " + r.getWidth());
7.         System.out.println("R-Area: " + r.area());
8.         r.setWidth(9.0); r.setHeight(7.0);
9.         Shape s = r; System.out.println("S-Area: " + s.area());
10.        System.out.println("S-Point: " + s.getPoint().getX() + ", " + s.getPoint().getY());
11.        p.setX(5); p.setY(7);
12.        Circle c = new Circle(p, 7.0);
13.        System.out.println("C: " + c.getRadius() + ", C-Area: " + c.area());
14.        c.setRadius(5.0);
15.        s = c; System.out.println("S-Area: " + s.area());
16.    }
17. }

```

Output of this program:

```

Point: 11, 22
R: 50.0, 30.0
R-Area: 1500.0
S-Area: 63.0
S-Point: 11, 22
C: 7.0, C-Area: 153.86
S-Area: 78.5

```

- Q6 a) What happens if a class partially implements an interface? Is it possible for an interface to extend another two interfaces in Java? 3+2

- b) Write a program to create two interface Testing1 and Testing2. In Testing1 interface you have two methods method1() and method2() and in Testing2 interface you have two methods methodA() and methodB(). Implements all these interfaces in another class named MyClass. 6

- c) Using either option create a class 'TimeThread' which will print "Thread started." at the beginning and print the seconds like '1 2 3 ... 60' and after 60 second it'll print "Thread stopped." Create and start the thread from "ThreadDemo" class. 7

- d) Write a code using Java threading where a Deadlock condition must occur. 7